



Pentahelix-Based Community Empowerment of Smoked *Sardinella* Microenterprises in Coastal Southeast Sulawesi, Indonesia

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Abstract: A community empowerment and local enterprise development model for smoked *Sardinella* microenterprises has been developed to investigate their operation in coastal Southeast Sulawesi, Indonesia. Smoked fish processing is a household livelihood activity that preserves local knowledge and supports coastal income, but producers involved in the industry still face unstable supply of raw materials, traditional equipment, simple packaging, limited chances of certification, restricted market access, weak business organization, and fragmented institutional support. A qualitative case study was conducted from September to December 2025 in Laeya and South Palangga districts, South Konawe Regency. Data were collected through in-depth interviews, participatory observation, document review, and focus group discussions involving 102 informants, including 42 producers and 60 Pentahelix stakeholders from the community, government, academia, business, and media groups. The data were then analyzed using five approaches: the Input-Process-Output-Outcome-Impact framework; Strengths, Weaknesses, Opportunities, and Threats (SWOT) analysis; the Internal Factor Analysis Summary (IFAS); the External Factor Analysis Summary (EFAS); and thematic interpretation. The internal score of 2.70 and external score of 2.65 placed the enterprises in a growth-oriented Strength-Opportunity position. The findings revealed that producers had strong traditional skills, local product identity, social trust, and regular local demand. However, their empowerment remained functional rather than transformative because production, market access, and institutional capacity were still weak. The novelty of the study lies in specifying how bonding social capital could be converted into bridging and linking mechanisms through a community-centered Pentahelix arrangement. The model offered policy guidance for shared processing facilities, certification pathways, group-based finance, and digital market linkage. These findings contribute to community development scholarship by clarifying the mechanism through which local enterprise assets could move from functional survival to transformative empowerment.

Keywords: Community empowerment; Pentahelix collaboration; Small-scale fisheries; Smoked *Sardinella*; Microenterprises; Social capital; Strengths, Weaknesses, Opportunities and Threats analysis

1. Introduction

Small-scale fisheries support food security, employment, household income, and social resilience in many coastal communities (Martuscelli, 2022). In coastal Indonesia, post-harvest fish processing also preserves local knowledge and creates value from highly perishable aquatic resources. One of the most common processing methods is preservation using smoke from burning wood, because it improves flavor, extends shelf life, and enables households to sell fish beyond the moment of landing (Belton et al., 2022). Post-harvest handling is also important because processed fish products are closely connected with food systems, livelihoods, and local market continuity in many coastal communities. These functions are important for small pelagic fish, whose availability

changes with season, weather, and fishing conditions.

The study focused on smoked *Sardinella* microenterprises in South Konawe Regency, Southeast Sulawesi. *Sardinella* sp., locally known as ikan tembang, is an important raw material for coastal processors. According to unpublished administrative landing records obtained from the Torobulu Fish Auction Place, total fish landings reached 446,840 kg in 2024, of which approximately 214,000 kg was *Sardinella*. This volume illustrated that *Sardinella* is a strategic local commodity for value-added processing and coastal household livelihoods. The study by Siang et al. (2023) supported the broader argument that the availability of raw materials was an important condition for sustainable fisheries processing in micro and small enterprises in nearby Kendari, Indonesia.

Despite this potential, producers of smoked *Sardinella* still operate under several constraints. Production relies on traditional smoking equipment, manual labor, and household-based management. Packaging remains simple and rarely includes brand identity, product information, certification, or shelf-life guidance. Marketing depends on collectors, retailers, local markets, and regular customers. Digital promotion and wider distribution remain limited, so the product has not been fully moved from survival-level household production to a competitive local enterprise (Mappirewa et al., 2024).

Previous studies noted that development of small-scale fisheries depended on product quality, post-harvest handling, market access, and institutional support (March & Failler, 2022). Product differentiation and customer value also influence the competitiveness of enterprises, especially when local products compete in wider markets (Saeng-on et al., 2022). Studies on smoked fish enterprises in Indonesia and other coastal regions further commented that technology, packaging, hygiene, and value-chain coordination affected product quality and market growth (Wicaksana et al., 2022). These findings indicated that technical improvement was important though it could not stand alone.

Community-based enterprises are shaped by social relations, local norms, trust, family labor, and informal cooperation. Social capital could strengthen collective action and reduce transaction costs when small producers face uncertain supply, limited capital, and weak formal institutions (Saz-Gil et al., 2021). In coastal fisheries, family-based and community-oriented enterprises also have to confront different social impacts from purely business-oriented models (Muis, 2022). Therefore, the development of smoked fish microenterprises should connect business strategies with community empowerment.

The broader significance of this case goes beyond South Konawe. Many coastal microenterprises in developing economies face the same upgrading dilemma: They possess local skills and strong bonding ties, but they lack bridging ties to markets and linking ties to institutions. By examining how a smoked fish microenterprise could be upgraded through Pentahelix collaboration, this paper contributed to international community development literature on asset-based livelihood transformation, collaborative governance, and social capital conversion in local enterprise systems.

The main research gap lies in limited integration between local enterprise development and community empowerment for a specific post-harvest fishery product. Many studies examined fish processing, packaging, market access, or empowerment as separate issues. Fewer studies explained how internal business capacity, social capital, and multi-actor collaboration could be integrated into a practical empowerment model. This gap is relevant to South Konawe because producers face simultaneous problems in production, continuity of raw materials, product standardization, capital, market access, and institutional coordination.

This study approached the issue by applying the Input-Process-Output-Outcome-Impact framework; Strengths, Weaknesses, Opportunities, Threats (SWOT) analysis; Internal Factor Analysis Summary (IFAS); External Factor Analysis Summary (EFAS); and Pentahelix collaboration to build an applied empowerment model. The main objective is to formulate business development strategies and construct a Pentahelix-based community empowerment model for smoked *Sardinella* microenterprises in coastal Southeast Sulawesi. The thesis of the study is that the sustainability of smoked *Sardinella* microenterprises depends on upgrading local strengths through technology adoption, producer organization, certification, market linkage, and coordinated support from community, government, academia, business, and media actors.

The proposed model differed from conventional Pentahelix applications because it placed the producer community as the central coordinating actor rather than a passive beneficiary. It also connected each actor's role to a specific empowerment mechanism: Bonding social capital supported daily production, bridging social capital supported market access, and linking social capital supported certification, finance, and policy access. This mechanism is expected to move empowerment from a functional stage, where producers merely survive, to a transformative stage, where producers gain collective bargaining power and institutional control.

2. Methodology

2.1 Study Area and Research Design

The study was conducted in South Konawe Regency, Southeast Sulawesi, Indonesia, from September to December 2025. The sites of the field study were Laeya and South Palangga districts. These districts were selected

purposively because both have active smoked *Sardinella* processing, direct links with local fish landing systems, and visible interaction among producers, traders, and local institutions. Laeya is connected to the Torobulu Fish Auction Place, while South Palangga has coastal landing points that supply fresh fish to household processors.

The design of a qualitative case study was adopted because the research examined a specific local enterprise within its social, economic, and institutional context. The qualitative design allowed the study to explore production practices, community relations, market channels, institutional roles, and empowerment needs in depth. It was strengthened by semi-quantitative strategic assessment through SWOT, IFAS, and EFAS.

2.2 Informants and Data Collection

Informants were selected purposively based on their involvement in smoked *Sardinella* processing, marketing, institutional support, and community empowerment. The total number of informants was 102. The business development analysis involved 42 active producers who directly handled raw material procurement, smoking, packaging, sales, and household-scale business management. The empowerment model involved 60 additional Pentahelix stakeholders representing community, government, academics, business actors, and media. The 60 Pentahelix stakeholders were separate from the 42 active producers, so the total number of informants remained 102.

The Pentahelix stakeholders consisted of 10 community representatives, 5 business actors, 35 government representatives, 5 academics, and 5 media actors. The 10 community representatives were local community members involved in family labor, neighborhood support, producer-group discussion, and community-based cooperation, but they were not counted among the 42 active producers. The business group consisted of five distributors or sellers involved in smoked *Sardinella* marketing. The government group consisted of 35 representatives from village, district, and regency government, including offices related to marine and fisheries, community and village empowerment, cooperatives and small enterprises, and industry and trade. The academic group consisted of five academic representatives with knowledge of fish processing, extension, and community development. The media group consisted of five local media actors or social media users involved in product promotion and public information.

Data were collected through in-depth interviews, participatory observation, document review, and focus group discussions. Interviews used a semi-structured guide covering production practices, raw material supply, technology, packaging, capital, marketing, institutional support, and empowerment needs. Observation was done at processing sites, fish landing points, distribution locations, and community areas. Documents included local fishery production data, village and district profiles, policy documents, program reports, microenterprise development records, and the unpublished 2024 fish landing record from the Torobulu Fish Auction Place. The Torobulu Fish Auction Place document was obtained during fieldwork in 2025 from the local auction-place record holder and contained annual commodity landing data for 2024. Because these records are not publicly accessible, they are reported here as contextual field information and are not included in the References. Focus group discussions were used to validate field findings, identify strategic factors, and prioritize the empowerment model.

2.3 Data Analysis

Data analysis was conducted in five stages as detailed below. First, field data were classified using the Input-Process-Output-Outcome-Impact framework. Input included human resources, supply of raw materials, equipment, capital, access to information, and training. Processes included smoking, packaging, distribution, marketing, institutional support, and community participation. Output included product quality, distribution channels, and social networks. Outcome included income, business independence, and technical capacity. Impact included community empowerment and household welfare.

Second, internal and external strategic factors were identified using SWOT analysis. Internal factors covered production capacity, product identity, skills, capital, technology, packaging, marketing, and institutions. External factors covered market demand, policy support, raw material seasonality, weather, competition, price fluctuation, and access to partners. Third, IFAS and EFAS were prepared to quantify the relative importance of strategic factors.

To improve analytical transparency, the weighting and rating procedure was done in three steps. First, the initial list of factors was derived from interview coding, observation notes, documents, and focus group discussions. Second, the 42 producers and 60 selected Pentahelix stakeholders discussed the relative importance of each factor during focus group sessions. Third, the research team normalized the agreed factor weights so that the total weight equaled 1.00 and assigned consensus ratings, based on participants' assessment of field conditions. Ratings used a four-point scale, where 1 indicated a major weakness or poor response, 2 indicated a minor weakness or below-average response, 3 indicated a minor strength or above-average response, and 4 indicated a major strength or strong response. Decimal ratings were used when focus group consensus produced an average between two categories.

The weighted score for each factor was calculated by multiplying its normalized weight by its rating. The total

IFAS and EFAS scores were then used to determine the strategic position of the enterprises. Weights, ratings, and weighted scores in the IFAS and EFAS matrices were rounded to two decimal places, and the total scores were calculated from the rounded values shown in the tables. Scores above 2.50 indicated that strengths and responses to opportunities were stronger than weaknesses and threats. This procedure enabled the strategic analysis to be replicable and allowed readers to evaluate how the final scores were produced.

Fourth, alternative strategies were formulated through Strength-Opportunity, Weakness-Opportunity, Strength-Threat, and Weakness-Threat combinations. These strategies were discussed in focus group discussions to assess urgency, relevance, feasibility, and potential impact. Fifth, thematic analysis was used to construct the Pentahelix-based empowerment model. Themes were organized around stakeholders' roles, collaboration patterns, capacity needs, institutional support, access to market, product quality, and community participation.

2.4 Validity and Ethical Considerations

Data validity was strengthened through source triangulation, method triangulation, and member checking. Source triangulation compared information from producers, distributors, community members, government officers, academics, business actors, and media actors. Method triangulation compared interview data with observation notes, documents, and focus group discussions. Member checking was done by confirming key findings with selected informants and participants in focus group discussions.

Before the collection of data, participants received information about the purpose of the study, the voluntary nature of participation, and the use of data for academic publication. Personal identifiers were not used in the analysis. The data were reported in aggregate form to ensure confidentiality.

3. Results

3.1 Characteristics of Smoked *Sardinella* Microenterprises

Findings from the field study demonstrated that smoked *Sardinella* processing in South Konawe is a household-scale enterprise based on local raw materials, family labor, inherited skills, and informal trading networks. Producers obtain fresh *Sardinella* from local landing points, fish collectors, or the Torobulu Fish Auction Place. According to unpublished administrative landing records reviewed during fieldwork at the Torobulu Fish Auction Place, *Sardinella* accounted for a substantial share of total fish landings in 2024. This indicates its local importance as a raw material for smoked-fish processing and value addition.

Supply increases during favorable fishing months and decreases during bright moon periods, strong winds, and rough seas. This fluctuation of supply affects the volume of production, the stability of cost, and household income. Similar supply-chain pressure in processed fish systems could generate constraints across trading, processing, and marketing relations (Agbekpurnu et al., 2025).

Most producers adopt traditional wood-fired stoves, manual smoking tools, bamboo or metal skewers, and household labor. Traditional practices maintain the distinctive flavor and identity of smoked *Sardinella*, but they also create variations in color, texture, dryness, and durability. Some producers implement minor improvements, such as replacing bamboo skewers with stainless steel. This finding indicates that innovation exists, but the application is limited and uneven. Improved smoking technology could increase processing efficiency and support product quality when it remained compatible with producers' capacity (Asamoah et al., 2025; Jayawickrama et al., 2022).

Marketing remains local and relationship based. Producers sell products to collectors, retailers, local markets, and regular consumers. Packaging is simple and often lacks labels, certification, brand identity, or standardized product information. Digital promotion is limited, as similar smoked fish businesses in South Sulawesi face growth constraints when packaging, promotion, and market linkage remain to be weak (Saediman et al., 2021). As a result, smoked *Sardinella* has stable local demand but weak market readiness for wider distribution. Institutional support remains fragmented because producer groups and formal partnerships prove inadequate.

3.2 Input-Process-Output-Outcome-Impact Analysis

The Input-Process-Output-Outcome-Impact analysis demonstrated that the availability of local skills and raw materials was the strongest input. Producers have practical knowledge in selecting fish, adjusting the duration of smoking, controlling fuel utilization, and maintaining product taste. Apparently, the input constraints include limited capital, traditional technology, weak access to training, and irregular supply of raw materials.

As shown in the process component, production is traditional and labor-intensive. Producers manage smoking and packaging based on experience, but the process lacks formal quality standards. This condition is important because bacterial contamination and control strategies play a central role in post-harvest fish handling (Kakerissa et al., 2024). Distribution channels are active, yet they are short and informal. The output includes distinctive

smoked fish products, regular local customers, and emerging social networks. However, these outputs have not produced strong market expansion or institutional upgrading.

The outcome component indicates that smoked *Sardinella* processing contributes to household income and local employment. It also reduces post-harvest loss by converting fresh fish into a higher-value product. However, income remains unstable because production depends on season, prices of raw materials, fuel costs, and local demand. The impact component suggests functional empowerment because producers can survive and continue their production. Yet, they have not achieved transformative empowerment through collective bargaining power, formal institutions, or stronger market control.

In this study, functional empowerment refers to the ability of producers to maintain daily production and household income with existing skills and resources. Transformative empowerment refers to a higher level of capacity in which producers could organize collectively, negotiate with markets and institutions, access certification and finance, and influence decisions that affect the environment of their enterprises. A growth-oriented strategy refers to a strategic position in which existing strengths could be used to capture external opportunities, while still managing risks caused by seasonality and uncertainty of the market.

3.3 Strategic Position and Development Priorities

SWOT analysis identified internal strengths, internal weaknesses, external opportunities, and external threats affecting smoked *Sardinella* microenterprises. The detailed IFAS calculation is presented in Table 1, which shows how internal factors are weighted, rated, and converted into weighted scores.

Table 1. Summary of internal factor analysis for smoked *Sardinella* microenterprises

Internal Factor	Type	Weight	Rating	Weighted Score
Traditional smoking skills	Strength	0.12	4.00	0.48
Distinctive taste and product identity	Strength	0.11	4.00	0.44
Availability of local raw materials	Strength	0.10	3.20	0.32
Social trust and regular customers	Strength	0.09	3.10	0.28
Household labor and inherited knowledge	Strength	0.08	3.00	0.24
Traditional equipment	Weakness	0.12	2.00	0.24
Simple packaging and labeling	Weakness	0.11	1.90	0.21
Limited certification and quality standards	Weakness	0.10	1.80	0.18
Weak capital access	Weakness	0.09	1.90	0.17
Low digital marketing and organization capacity	Weakness	0.08	1.75	0.14
Total		1.00		2.70

Table 1 presents the internal score of 2.70. The strongest internal assets are traditional smoking skills, product identity, access to raw materials, social trust, and inherited knowledge. The weakest internal factors are simple packaging, limited certification, traditional technology, weak access to capital, and low digital marketing capacity. Because the total internal score was above 2.50, the enterprise had stronger internal assets than internal constraints.

The external factors were calculated through EFAS, as shown in Table 2. The EFAS matrix clarifies the opportunities and threats that influence the upgrading potential of smoked *Sardinella* microenterprises.

Table 2. Summary of external factor analysis for smoked *Sardinella* microenterprises

External Factor	Type	Weight	Rating	Weighted Score
Stable local demand	Opportunity	0.12	3.50	0.42
Potential government support	Opportunity	0.11	3.30	0.36
Regional specialty market	Opportunity	0.10	3.40	0.34
Potential of producer collaboration	Opportunity	0.09	3.20	0.29
Appropriate technology and digital promotion	Opportunity	0.10	3.10	0.31
Weather uncertainty	Threat	0.12	1.90	0.23
Seasonal fish supply	Threat	0.10	1.80	0.18
Price fluctuation of raw materials	Threat	0.09	1.80	0.16
External product competition	Threat	0.08	1.70	0.14
Weak storage and shared facilities	Threat	0.09	2.40	0.22
Total		1.00		2.65

As listed in Table 2, the external score is 2.65. The most important opportunities are stable local demand, government support, regional specialty markets, producer collaboration, and access to appropriate technology. The main threats are weather uncertainty, seasonal supply, price fluctuation, external competition, and weak storage facilities. Since the external score is also above 2.50, the enterprises have sufficient opportunity to grow if risks are managed collectively.

The combined IFAS and EFAS scores place the enterprises in the Strength-Opportunity quadrant. This position supports a growth-oriented strategy that applies local strengths to capture market and institutional opportunities. However, the growth strategy should be accompanied by defensive measures because the enterprise is vulnerable to supply seasonality, fluctuation of prices, and weak infrastructure.

The strategic alternatives generated from the SWOT matrix are summarized in Table 3, which translates the strategic position into practical priorities for enterprise upgrading and community empowerment.

As summarized in Table 3, the Strengths-Opportunities (SO) strategy is prioritized as the main development strategy, whereas the Weaknesses-Opportunities (WO), Strengths-Threats (ST), and Weaknesses-Threats (WT) strategies mainly function as capacity-building and risk-reduction strategies. This interpretation clarifies that the growth-oriented position does not mean unregulated expansion. Growth should be organized through producer groups, certification pathways, shared facilities, and stronger links with market actors.

Table 3. Strategic alternatives and priority actions

Strategy	Strategic Logic	Priority Actions	Expected Effect
SO	Use strengths to capture opportunities	Develop local branding, improve packaging, expand regional markets, and promote smoked <i>Sardinella</i> as a local specialty	Move local identity into market value and strengthen producers' confidence
WO	Use opportunities to reduce weaknesses	Provide training, facilitate certification, improve bookkeeping, introduce digital promotion, and support appropriate smoking technology	Build technical and managerial capacity
ST	Use strengths to reduce threats	Coordinate raw material planning, diversify products, improve storage, and strengthen social networks for market continuity	Improve resilience under seasonal and price uncertainty
WT	Reduce weaknesses and avoid threats	Create producer groups, shared processing centers, collective purchasing, and access to group-based finance	Reduce vulnerability and strengthen collective bargaining power

Note: SO = Strengths-Opportunities; WO = Weaknesses-Opportunities; ST = Strengths-Threats; WT = Weaknesses-Threats.

3.4 Pentahelix Empowerment Model

The empowerment model is built around five actors: community, government, academics, business actors, and media. The proposed role distribution is presented in Table 4. The model remains a field-validated conceptual model rather than an already implemented intervention.

Table 4. Pentahelix empowerment model for smoked fish microenterprises

Stakeholder	Main Roles	Practical Contributions	Empowerment Mechanism
Community	Core producer and knowledge holder	Maintain production, local skills, cooperation, and product identity	Transform bonding capital into collective organization
Government	Regulator and facilitator	Support licensing, certification, equipment, training, data collection, and interagency coordination	Create linking capital and policy access
Academics	Knowledge and technology provider	Support hygienic processing, packaging innovation, quality control, business mentoring, and applied research	Convert local knowledge into improved practice
Business actors	Market and value-chain partner	Support distribution, market information, product standards, and business partnerships	Create bridging capital and market discipline
Media	Promotion and communication actor	Support digital promotion, product branding, consumer awareness, and regional product visibility	Build visibility, consumer trust, and market reach

Table 4 places the community as the core actor because producers possess the skills, labor, and local knowledge that sustain the enterprise. The other four actors provide enabling support. Government actors facilitate licensing, certification, equipment assistance, and coordination across agencies. Academics support research, training, technology transfer, product standardization, and business mentoring. Business actors connect producers with distribution networks, market information, and quality requirements. Media actors strengthen visibility, storytelling, and regional branding.

The model responded to the main weakness in the field. Producers already have bonding social capital within families and local networks, but they have weaker bridging and linking social capital with formal institutions, markets, and media. The Pentahelix model therefore works as a linking mechanism that connects local capacity with external resources. This connection is required to move producers from functional empowerment to

productive and transformative empowerment.

The model had not been empirically implemented as a pilot program during this study. Its validation was restricted to stakeholders' discussion and interpretation of field evidence. Therefore, the model should be treated as a practical framework for intervention design that requires future testing through pilot implementation, monitoring, and evaluation.

4. Discussion

The findings demonstrated that smoked *Sardinella* microenterprises were not weak because they lacked local capacity. They were constrained because local capacity had not been connected to technology, formal organization, certification, and wider markets. This distinction is important for community development because it shifts the intervention logic from assistance-based support to asset-based upgrading. Producers should not be treated only as beneficiaries. They should be treated as strategic actors who already control valuable skills, product identity, and local networks. Studies on household food security in coastal communities also indicated that social capital was linked to livelihood resilience (Overå et al., 2022).

This interpretation is consistent with resource-based thinking and findings on coastal women empowerment in small-scale fisheries (Freeman & Svells, 2022). Local taste, inherited smoking skills, and product identity could function as valuable resources when they are supported by packaging, quality consistency, and market communication. Barney (1991) explained that valuable, rare, and difficult-to-imitate resources could form the basis of sustained competitive advantage. Moreover, Sari & Nissa' (2025) concluded that women-managed fish processing enterprises faced structural constraints that influenced competitive advantages in coastal contexts. In this study, resources existed in local enterprises, but they required institutional and market support before they could become a competitive advantage.

The study extended social capital theory by showing that bonding social capital was necessary but insufficient for enterprise transformation. Bonding ties help producers maintain production and trust, but they do not automatically generate certification, finance, technology, or wider market access. Bridging ties with business actors and linking ties with government, training providers, and media are required to convert local solidarity into institutional and market capacity (Woolcock & Narayan, 2000). This finding refined social capital theory in the context of coastal microenterprises by clarifying that empowerment depended on the conversion among bonding, bridging, and linking capital rather than on the presence of trust alone.

The findings also contributed to community development theory by demarcating the differences between functional and transformative empowerment. Functional empowerment allows producers to survive, produce, and maintain household income. Transformative empowerment requires collective agency, access to rules and institutions, and the capacity to negotiate market relations. This distinction is consistent with empowerment theory, which emphasizes people's capacity to gain control over resources, decisions, and conditions affecting their lives (Zimmerman, 1995). In the context of small-scale fisheries, empowerment also requires collective agency, institutional access, and stronger participation in value-chain relations (Freeman & Svells, 2022). The study therefore added a concrete enterprise-based pathway for explaining how community empowerment could progress from individual livelihood resilience to collective economic agency.

The Pentahelix model contributes to collaborative governance and Pentahelix-based development frameworks. Existing Pentahelix models often describe the presence of five actors but do not always specify the mechanism by which each actor changes the empowerment status of a community. This study specified those mechanisms: Government built linking access, academics translated knowledge into technology and mentoring, business actors built market bridges, media built visibility, and community actors retained ownership over the enterprise. This arrangement offered an advantage over top-down empowerment approaches because it protected community agencies, while mobilizing external support (Ansell & Gash, 2008; Carayannis & Campbell, 2010).

The strategic position in the Strength-Opportunity quadrant has practical implications. Development programs should prioritize growth strategies, but growth must be organized and risk aware. Product branding, improved packaging, certification, and digital promotion could expand markets. Meanwhile, seasonal supply of raw materials, weather uncertainty, and price fluctuation require adaptive production planning, collective procurement, and shared processing facilities. This combination prevents the growth strategy from becoming overly market-oriented, while ignoring producer vulnerability.

The mechanism through which the Pentahelix model could move producers from functional to transformative empowerment consisted of four steps. First, producers should be organized into groups or cooperatives to convert individual capacity into collective capacity. Second, shared smoked-fish processing centers should be established to improve hygiene, access to technology, consistency of quality, and storage. Third, certification and product standardization programs should be implemented to meet market requirements. Fourth, market linkage should be strengthened through business partnerships and community-based digital marketing platforms. These steps connected social capital, technical upgrading, institutional access, and market expansion into one empowerment pathway.

Compared with studies that focused primarily on fish processing technology, marketing practices, or market strategies in smoked fish, this study offered a broader community development contribution (Abror et al., 2022). Processing technology improves product quality, but empowerment requires capacity building, institutional access, and control over market relations. Compared with studies that focus mainly on empowerment, this study provided a more operational pathway by identifying the enterprise position through SWOT and translating it into Pentahelix roles. Although much remains to be done, the main application of this study is a practical model that local government and community facilitators could use to organize support for smoked fish microenterprises.

The policy implications are specific. Local government should establish shared smoked-fish processing centers equipped with hygienic smoking facilities, storage space, and packaging equipment. Technical agencies should implement certification and product standardization programs for fisheries microenterprises, including food safety, labeling, halal certification, and business licensing. Cooperative and microfinance agencies should develop group-based financing schemes for producers rather than relying only on individual borrowing. Community facilitators and media actors should develop community-based digital marketing platforms that promote the identity of local product, traceability, and consumer trust.

This study has several limitations. First, the investigation used a qualitative case study design, so the findings were context-specific and should be transferred cautiously to other coastal regions. Second, the study focused on social, institutional, and strategic aspects. It did not include laboratory testing of product safety, shelf life, or nutritional quality. Third, the proposed Pentahelix model was based on field interpretation and stakeholders' validation. To acknowledge the limitations of the current research design, future studies should test the model through pilot implementation, longitudinal monitoring, and comparative research across coastal communities.

5. Conclusions

This study concluded that smoked *Sardinella* microenterprises in coastal Southeast Sulawesi had strong developmental potential because they were supported by local raw materials, inherited smoking skills, distinctive product taste, social trust, and regular local demand. However, the enterprises remained constrained by traditional technology, simple packaging, limited quality standardization, weak certification, narrow access to the market, limited capital, and informal institutions. These conditions revealed that empowerment had reached a functional level, but it had not yet become transformative.

The strategic analysis placed the enterprises in a Strength-Opportunity position with an internal score of 2.70 and an external score of 2.65. This conclusion was supported by transparent IFAS and EFAS matrices that explored the weight, rating, and weighted score of each strategic factor. The most relevant research direction was a growth-oriented strategy that upgraded product quality, packaging, labeling, certification, access to the market, and producer organization without removing local identity.

The study pointed to new possibilities that the transition from functional to transformative empowerment required the conversion of bonding social capital into bridging and linking mechanisms through a community-centered Pentahelix model. The model differed from conventional multi-actor collaboration because it kept producers as the core actors, while assigning specific enabling roles to government, academics, business actors, and media.

The scientific contribution of the study lies in refining community development and social capital theory in a coastal enterprise context. Policy recommendations should prioritize shared smoked-fish processing centers, certification and standardization programs, group-based financing, producer cooperatives, and community-based digital marketing platforms. These recommendations could strengthen coastal livelihoods, product competitiveness, and institutional capacity. Future research should test the proposed model through pilot programs, evaluate product safety and shelf life, and compare similar fish processing microenterprises in other coastal areas.

Author Contributions

Conceptualization, W.O.H. and U.R.; methodology, W.O.H., U.R., and D.S.; validation, U.R., D.S., and S.A.; formal analysis, W.O.H.; investigation, W.O.H.; resources, W.O.H. and S.A.; data curation, W.O.H.; writing original draft preparation, W.O.H.; writing review and editing, W.O.H., U.R., D.S., and S.A.; visualization, W.O.H. and S.A.; supervision, U.R., D.S., and S.A.; project administration, W.O.H. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

Informed consent was obtained from all participants involved in the study. Participation was voluntary, and informants were informed that their responses would only be used for academic research and reported in aggregate form.

Data Availability

The qualitative data supporting the research findings are under privacy and ethical restrictions. The data are available from the corresponding author upon reasonable request for researchers who meet the criteria for accessing confidential data.

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Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

DeepL was used to assist with English translation and language refinement. The authors reviewed, revised, and verified all content, data interpretation, and references of the manuscript. DeepL was not used to fabricate data, results, or references.

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